

# Retail Sales Analytics and Customer Purchasing Behaviour Analysis

## CIND820 – Big Data Analytics Project

### Milestone 3: Initials Result and Coding

**Student Name:** Prabhjot Kaur

**Student Number:** 501409563

**Supervisor:** Tamer Abdou

**Date:** July 6, 2026



# Table of Contents

---

|                                           |       |
|-------------------------------------------|-------|
| 1. Introduction .....                     | 5     |
| 2. Data Analysis and Preparation.....     | 5-6   |
| 3. Model Development and Evaluation ..... | 7-9   |
| 4. Validation Protocol.....               | 10    |
| 5. Initial Business Findings.....         | 10    |
| 6. Limitations.....                       | 11    |
| 7. Conclusion.....                        | 11    |
| 8. References.....                        | 11    |
| 9.Appendix.....                           | 11-12 |

# 1. Introduction

---

This milestone presents the initial implementation and evaluation of the proposed customer analytics pipeline developed in Milestone 2. The objective is to demonstrate that the complete analytical workflow functions successfully, from data loading and preprocessing to customer segmentation using the Recency–Frequency–Monetary (RFM) framework. Rather than proposing a theoretical solution, this milestone provides a proof of concept by executing the pipeline and generating interpretable customer insights.

The analysis uses the Online Retail dataset containing transactional records from a UK-based online retailer between December 2010 and December 2011. Following data cleaning and feature engineering, customer-level behavioural variables were created to calculate Recency, Frequency, and Monetary values. These variables were used to build a baseline RFM segmentation model capable of identifying meaningful customer groups based on purchasing behaviour.

The implementation addresses the project's primary research objective of determining whether customer purchasing behaviour can be effectively segmented using RFM analysis to support customer retention, targeted marketing, and business decision-making. The results presented in this report demonstrate the successful execution of the proposed analytical pipeline and provide an initial evaluation of its effectiveness.

## 2. Data Analysis and Preparation

---

### 2.1 Data Overview

The analysis was conducted using the **Online Retail** dataset, which contains transactional records from a UK-based online retailer between **December 2010 and December 2011**. The dataset consists of customer purchases across multiple countries and includes transaction details such as invoice number, product description, quantity purchased, invoice date, unit price, customer identifier, and country. These variables provide the necessary information to examine customer purchasing behaviour and develop customer-level features for RFM analysis.

## **2.2 Data Preparation and Preprocessing**

Before model development, the dataset underwent a structured preprocessing pipeline to improve data quality and ensure reliable analytical results. Duplicate records were removed, cancelled transactions were excluded, and only valid transactions with positive quantities and positive unit prices were retained. Invoice dates were converted into a standard datetime format to enable time-based calculations. Missing Customer IDs were excluded from customer-level analysis because customer identification is essential for calculating Recency, Frequency, and Monetary values.

Feature engineering was then performed by calculating transaction-level revenue as the product of quantity and unit price. A monthly transaction variable was also created to support exploratory analysis and identify purchasing trends over time. Finally, a customer-level dataset was generated by aggregating transactions according to Customer ID, forming the foundation for the RFM model.

## **2.3 Descriptive Analysis**

Descriptive statistics and visualizations were generated to understand the characteristics of the cleaned dataset before customer segmentation. Histograms and boxplots were produced for the Recency, Frequency, and Monetary variables to examine their distributions, identify skewness, and detect potential outliers.

The analysis showed that all three variables exhibit positively skewed distributions, with a relatively small number of customers accounting for substantially higher purchasing frequency and spending. Boxplots confirmed the presence of high-value outliers, which represent customers with exceptional purchasing behaviour rather than data errors. These findings support the suitability of RFM analysis, as the methodology is specifically designed to distinguish customers according to differences in purchasing recency, frequency, and monetary value.

## **2.4 Evidence of Successful Execution**

The successful execution of the preprocessing pipeline was verified through the generated notebook outputs. The cleaned transactional dataset, customer-level dataset, engineered features, descriptive statistics, and visualizations confirmed that the analytical workflow executed correctly. These processed data were subsequently used to construct the RFM scoring model and customer segmentation analysis, demonstrating that the proposed pipeline is fully operational.

## 3. Model Development and Evaluation

---

### 3.1 Baseline Model Development

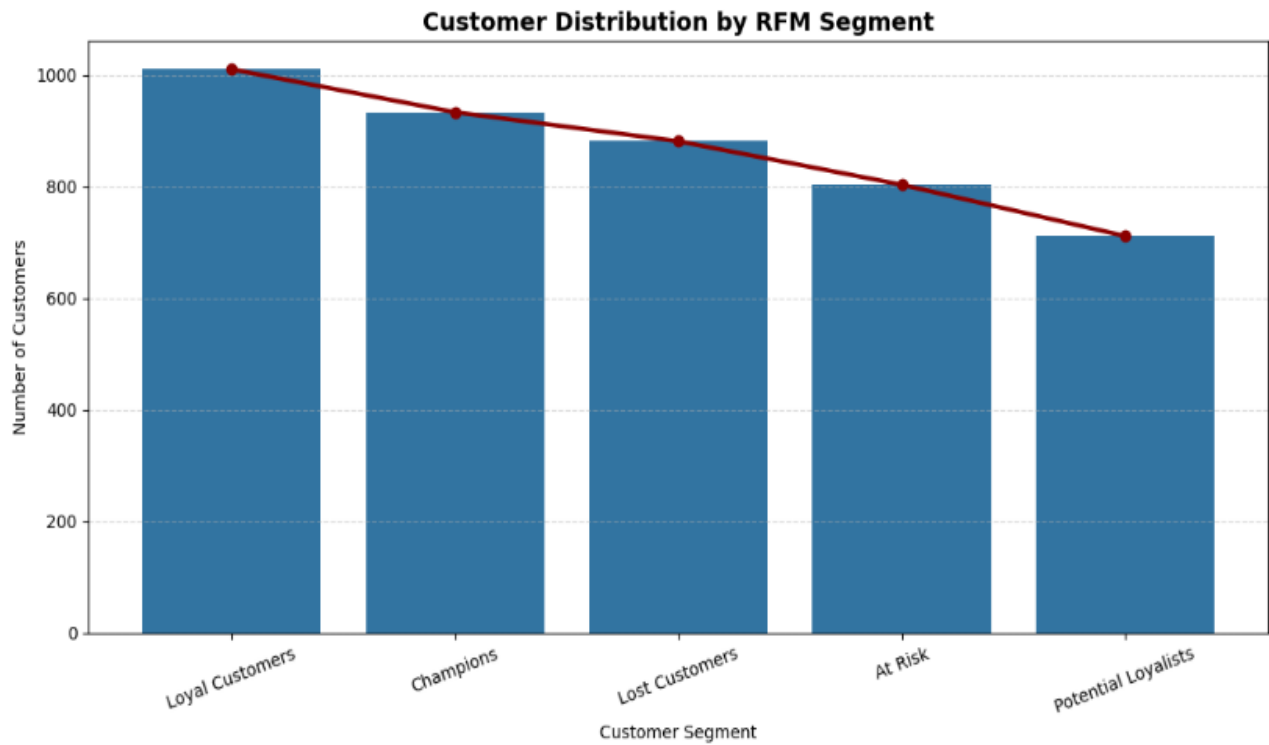
The proposed RFM methodology from Milestone 2 was implemented as the baseline customer segmentation model. Customer-level Recency, Frequency, and Monetary (RFM) variables were calculated from the cleaned transactional dataset. Recency was defined as the number of days since each customer's most recent purchase, Frequency as the number of unique purchase transactions, and Monetary as the total revenue generated by each customer during the observation period.

To ensure consistent comparison across customers, each RFM variable was converted into quintile-based scores ranging from 1 to 5. Customers with more recent purchases received higher Recency scores, while customers with higher purchasing frequency and greater monetary value received higher Frequency and Monetary scores. These standardized scores were combined to produce an overall RFM score for each customer, providing the basis for behavioural segmentation.

### 3.2 Customer Segmentation Results

Based on the calculated RFM scores, customers were grouped into five behavioural segments: **Champions**, **Loyal Customers**, **Potential Loyalists**, **At Risk**, and **Lost Customers**. These segments represent different levels of customer engagement and purchasing behaviour, allowing the retailer to distinguish between highly valuable customers and those requiring retention or reactivation strategies.

The resulting customer distribution demonstrates that the implemented RFM model successfully differentiated customers according to their purchasing behaviour. The segment distribution and supporting visualizations provide evidence that the proposed segmentation approach produces meaningful and interpretable customer groups suitable for business decision-making.

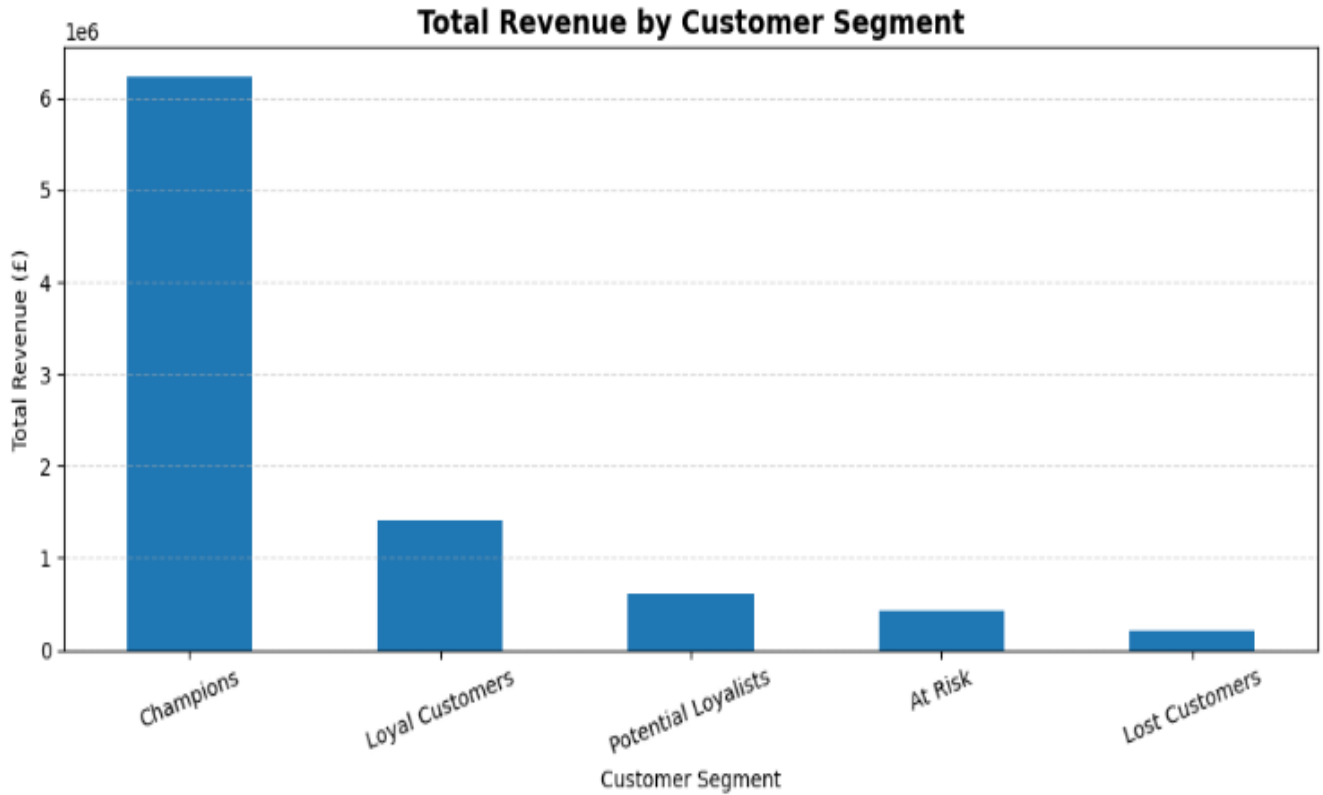


**Figure 1. Customer Distribution by RFM Segment**

### 3.3 Initial Model Evaluation

The effectiveness of the RFM model was evaluated by examining customer behaviour across the generated segments. Segment-level summaries showed clear differences in Recency, Frequency, and Monetary values, confirming that customers with higher RFM scores generally purchased more recently, purchased more frequently, and generated higher revenue than lower-scoring customers.

Revenue analysis further demonstrated that the **Champions** and **Loyal Customers** segments contributed a substantial proportion of total customer revenue, while **At Risk** and **Lost Customers** generated comparatively lower revenue. These results indicate that the implemented RFM framework successfully identifies high-value customer groups while simultaneously highlighting opportunities for customer retention and re-engagement.



**Figure 2. Total Revenue by Customer Segment**

### **3.4 Interpretation and Current Limitations**

The initial implementation confirms that the proposed RFM model functions as intended and produces interpretable customer segments aligned with the project objectives. Although the segmentation provides valuable behavioural insights, it is based solely on historical transactional information and does not incorporate demographic variables, product preferences, or future purchasing behaviour. Consequently, the current model should be viewed as an initial proof of concept that will be further refined and evaluated in the final stage of the project.

## 4. Validation Protocol

---

### 4.1 Validation Strategy

The implemented RFM model is a descriptive customer segmentation technique rather than a supervised machine learning model. Therefore, a traditional train-test split was not required because the objective was to summarize historical customer purchasing behaviour instead of predicting future outcomes. The analysis used the complete cleaned customer dataset to calculate Recency, Frequency, and Monetary values, ensuring that all available customer transaction history contributed to the segmentation process.

### 4.2 Data Integrity and Leakage Prevention

Data leakage occurs when information unavailable at the time of analysis is unintentionally incorporated into model development. In this project, all RFM variables were calculated exclusively from historical transaction records contained within the study period. No future transactions or external information were used during feature engineering or customer scoring. As a result, the segmentation process remained independent of future customer behaviour and maintained the integrity of the analytical pipeline.

### 4.3 Evidence of Functional Pipeline

The successful execution of the complete analytical workflow was verified through the Jupyter Notebook outputs. The pipeline successfully progressed from data ingestion and preprocessing to feature engineering, customer-level dataset creation, RFM scoring, customer segmentation, and model evaluation without execution errors. The generated tables and visualizations confirm that the proposed methodology is technically feasible and produces interpretable customer insights suitable for business decision-making.

## 5. Initial Business Findings

---

The initial results demonstrate that customer purchasing behaviour can be effectively segmented using the RFM framework. The identified customer groups provide actionable insights that support data-driven marketing and customer relationship management. High-value customers, including Champions and Loyal Customers, can be targeted through loyalty programs, personalized promotions, and customer retention initiatives. Potential Loyalists represent opportunities for future growth through targeted engagement strategies, while At Risk and Lost Customers can be addressed through reactivation campaigns designed to reduce customer attrition. Overall, the initial findings confirm that the implemented RFM model provides practical business value and directly addresses the project's research objectives.



## 6. Limitations

---

Although the initial implementation successfully demonstrates the feasibility of the proposed analytical pipeline, several limitations should be acknowledged. The analysis is based on a single historical dataset collected from one UK online retailer and therefore may not fully represent purchasing behaviour across different industries or time periods. Additionally, the RFM model considers only transactional behaviour and does not incorporate demographic, behavioural, or external market variables that may further improve customer segmentation. These limitations provide opportunities for future model enhancement and more comprehensive customer analytics.

## 7. Conclusion

---

This milestone successfully demonstrated the implementation of the proposed customer analytics pipeline from data ingestion to initial model evaluation. Data preprocessing, feature engineering, descriptive analysis, and RFM-based customer segmentation were successfully executed, producing meaningful customer groups that align with the project's research objectives. The initial results confirm that the RFM framework provides an effective baseline approach for analysing customer purchasing behaviour and generating actionable business insights. The completed proof of concept establishes a strong foundation for the final project, where the model will be further evaluated and translated into detailed business recommendations.

## 8. References

---

- Microsoft. (n.d.). [Introduction to GitHub](https://learn.microsoft.com/en-us/training/modules/introduction-to-github/). Microsoft Learn. <https://learn.microsoft.com/en-us/training/modules/introduction-to-github/>
- Toronto Metropolitan University. (n.d.). [Citations and style guides](https://library.torontomu.ca/guides/style/). <https://library.torontomu.ca/guides/style/>
- Toronto Metropolitan University. (2026a). [Academic integrity Policy 60](https://www.torontomu.ca/senate/course-outline-policies/academic-integrity-policy-60/). <https://www.torontomu.ca/senate/course-outline-policies/academic-integrity-policy-60/>
- Toronto Metropolitan University. (2026b). *CIND110: Data organization for data analysts*.
- Toronto Metropolitan University. (2026c). *CIND123: Data analytics: Basic methods*.
- Yang, A. X. (2004). How to develop new approaches to RFM segmentation. *Journal of Targeting, Measurement and Analysis for Marketing*, 13(1), 50–60. <https://doi.org/10.1057/palgrave.jt.5740131>

Christy, A. J., Umamakeswari, A., Priyatharsini, L., & Neyaa, A. (2021). RFM ranking—An effective approach to customer segmentation. *Journal of King Saud University – Computer and Information Sciences*, 33(10), 1251–1257. <https://doi.org/10.1016/j.jksuci.2018.09.004>

Chen, D., Sain, S. L., & Guo, K. (2012). Data mining for the online retail industry: A case study of RFM model-based customer segmentation using data mining. *Journal of Database Marketing & Customer Strategy Management*, 19(3), 197–208. <https://doi.org/10.1057/dbm.2012.17>

## 9. AI Coding Assistance Declaration

---

I used generative AI tools to assist with code development, debugging, documentation, and writing refinement throughout this milestone. I independently reviewed, tested, verified, and modified all generated code, analytical outputs, visualizations, and written content to ensure their accuracy, consistency with the project methodology, and compliance with the assignment requirements.

### GitHub Repository

The complete project repository, including the Jupyter Notebook, compiled report, project documentation, dataset (or dataset link), README file, and supporting materials, is available at:

[https://github.com/Prabhjot-kaur-tech/CIND820\\_Retail\\_sales\\_Analytics](https://github.com/Prabhjot-kaur-tech/CIND820_Retail_sales_Analytics)